Chapter 2 Modern Business Environment

1 Throughput Accounting Ratio

$$\frac{\text{Throughput per Bottleneck Minute}}{\text{Factory Cost per Bottleneck Minute}} \quad \text{OR} \quad \frac{\text{Time Required}}{\text{Time Available}}$$ (Highest TA ratio is constraint)

Chapter 3 Lean System & Innovation

1 Back Flushing (A/C System of JIT)

$$\left(\text{No. of units produced} \times \text{Unit count listed in bill of material for each component}\right) + \text{Adjustment for scrap during production process} = \text{No. of raw material units removed from stock}$$

2 Number of KANBAN

$$\text{(Boxes required for holding kanban size)} \quad \frac{\text{Quantity of Safety Stock} \quad \text{Consumption during lead period}}{\text{Kanban Size}}$$

3 TAKT – TIME

$$\frac{\text{Available Production Time}}{\text{Total Quantity Required}}$$

4 Formula's for Availability, Performance Quality Ratios (Newly introduced)

(I) Availability Ratio =

$$\frac{\text{Planned Production Time - Total time "Loss" Per Shift}}{\text{Planned Production Time}}$$

OR

$$\frac{\text{Operating time}}{\text{Planned Production Time}}$$

(II) Performance Ratio =

$$\frac{\text{Standard time required}}{\text{Actual time Taken}} \Big/ \frac{\text{Ideal cycle time x total count}}{\text{Operating time}}$$

OR

$$\text{Performance Ratio} = \frac{\text{Actual Production x Standard time p.u.}}{\text{Operating Time}}$$

(III) Quality Ratio =

$$\frac{\text{Actual Production - Quality loss due to scrap etc.}}{\text{Actual Productioin}} \Big/ \frac{\text{Good Count}}{\text{Total count}}$$

Chapter 6 Decision Making

1 Sales

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}}$$

(-) Variable Cost = Contribution

(-) Fixed Cost = Net Profit

$$\text{BEP (Point of No profit No Loss)} = \frac{\text{Fixed Cost}}{\left(\frac{\text{Contribution p.u.}}{\text{PV Ratio}}\ \text{OR}\right)}$$

$$\text{MOS (Point Beyond BEP to Earn Profit)} = \frac{\text{Net Profit}}{\left(\frac{\text{Contribution p.u}}{\text{P/V Ratio}}\ \text{OR}\right)}$$

$$\text{PV Ratio} = \frac{\text{Contribution p.u}}{\text{Sales Price p.u.}} \times 100$$

2 Key Factor / Limiting Factor (LF)

$$\text{LF} = \frac{\text{Contribution per unit}}{\text{Limiting Factor per unit}}$$

(Material / Machine Hrs/ Labour Hrs.)

3 Cost Indifference Point

$$\frac{\text{Change in Fixed Cost}}{\text{Change in Variable Cost}}$$

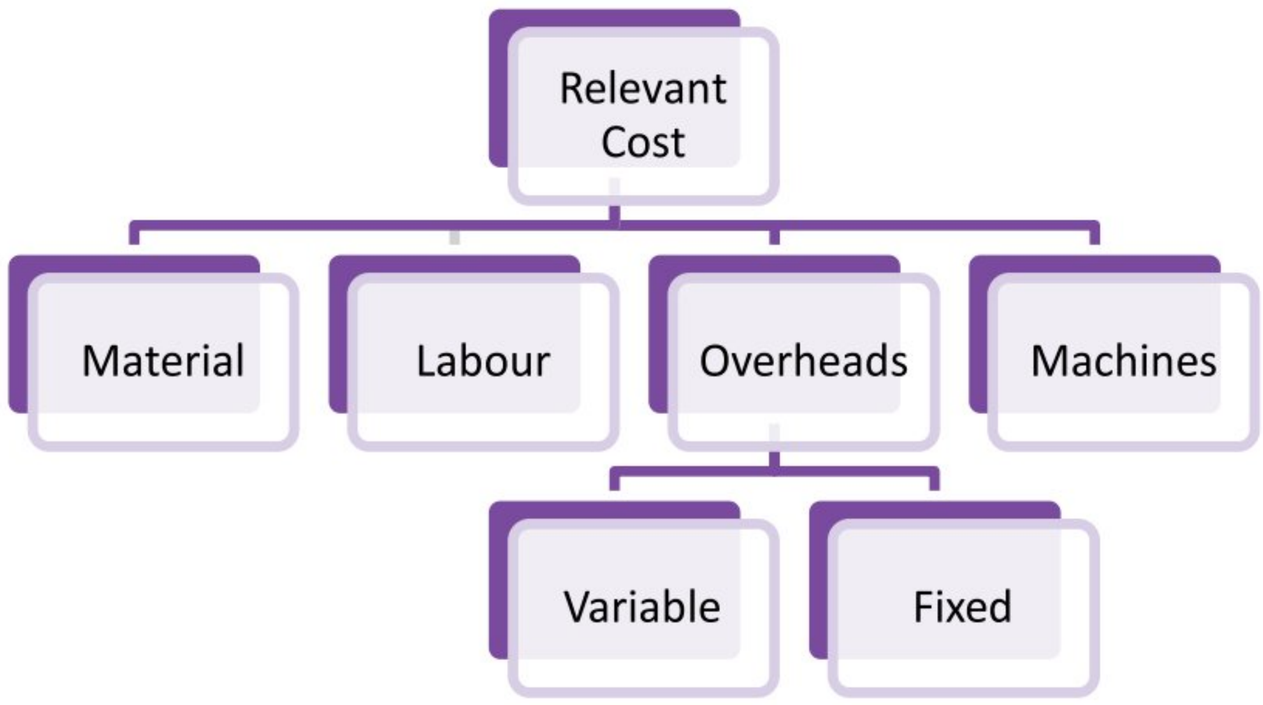
4 Jumping Break Even Point (Multiple BEP)

$$\frac{\text{Fixed Cost + Semi Variable Cost}}{\text{Contribution/ Unit}}$$

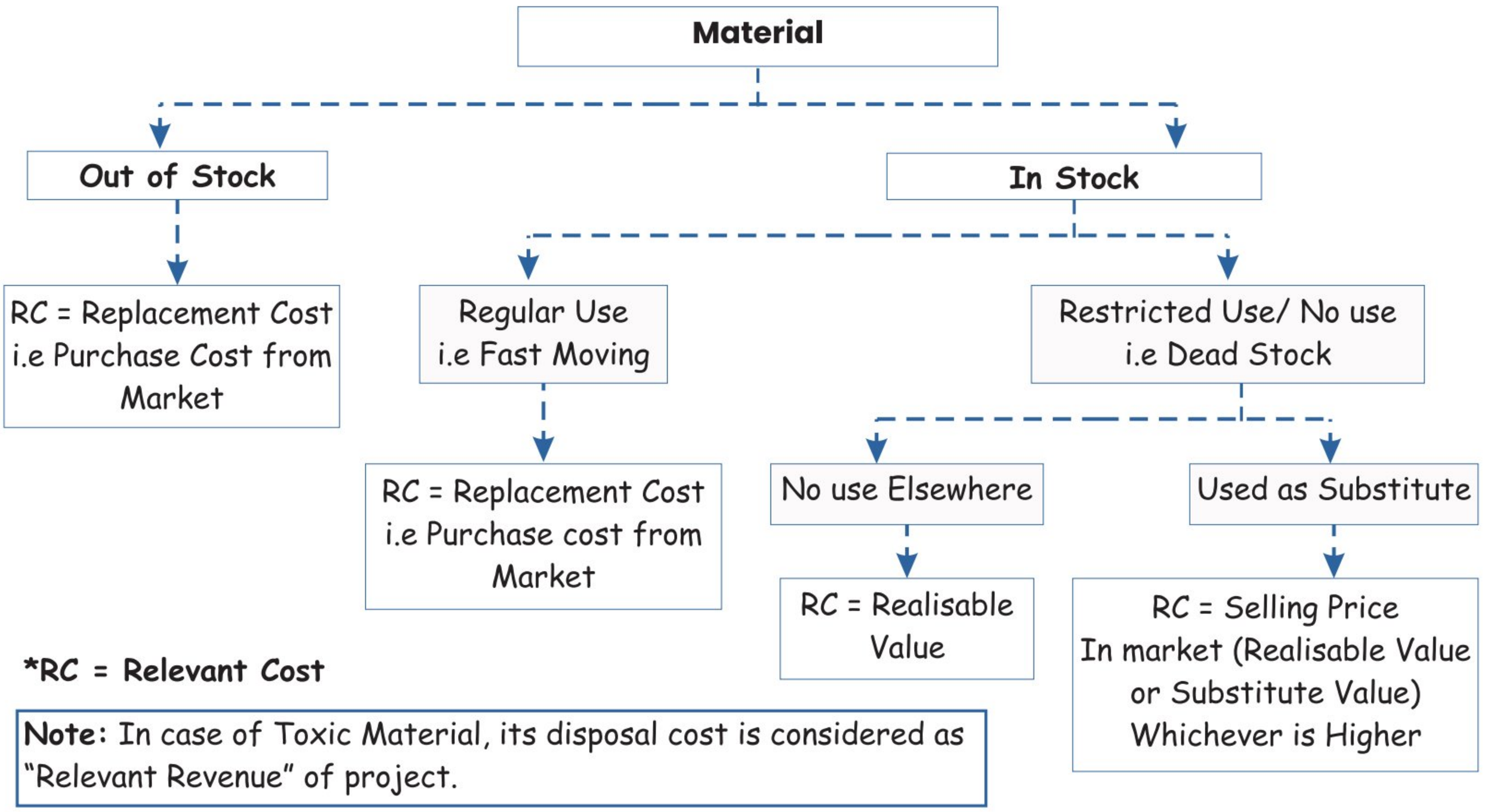
5 Shut Down Point / Shut Down Decision

Shut down point (in value)	$\frac{\text{Present F.C - Fixed cost if plant is shut down}}{\text{P/V Ratio}}$
Shut down point (in units)	$\frac{\text{Present F.C - Fixed cost if plant is shut down}}{\text{Contribution per unit}}$

6 Relevant Cost



7 Relevant cost of Materials



***RC = Relevant Cost**

Note: In case of Toxic Material, its disposal cost is considered as "Relevant Revenue" of project.

8 Relevant cost of Labour

Labour

- **Casual**
  - **Normal** → RC = Cost to be incurred For that Labour
  - **Busy** → RC = Cost to be incurred For that labour + contribution lost
- **Permanent**
  - **Normal** → Idle → RC = Nil (Since the company has spare capacity)
  - **Busy** → RC = Only contribution Lost (Normal cost will always be incurred)

9 Relevant cost of Overhead

Overheads

- **Variable** → RC = Variable Cost p.u. x No. of Units
- **Fixed**
  - **Avoidable** → RC = Additional Fixed Cost to be incurred
  - Committed Fixed Cost → RC = Nil
  - **Unavaidable / Absorbed** → Are Irrelevant because irrespective of the decision, they have to be Incurred

10 Relevant cost of Machinery

Machinery

- If the Project does not consume substantial part of machinery → RC = Irrelevant = Nil
- If the project consumes substantial part of machinery
  - **Regular Use** → RC = Difference of Replacement Cost
  - **No use of Machine** → RC = Difference of Realisable value

Chapter 8 Performance Measurement & Evaluation

1 Financial Performance Measures

Return on Investment

- Divisional profit as a % of assets. Measures gain/loss generated by initial investment

Used to influence financial decisions, compare company's profitability & analyze investments

The Formula for calculating ROI:

$$ROI = \frac{NPAT}{CE}$$

Where : NPAT = Net Profit After Taxes
CE = Capital Employed

Points to be kept in mind:

Instead of focusing on absolute size of divisions profit, company focuses on ROI
- Provides relative profitability.
- Useful for divisions of different sizes.

Limitation of ROI
- Against goal congruence & promotes individual profitability instead of company's profitability.

Residual Income

Used to overcome disadvantages of ROI

Amount left over operating profit after paying all cost of capital used to generate revenues

The Formula for calculating RI:

$$RI = OI - OE - \left(\text{min. reqd. return} \times \text{operating assets}\right)$$

Where : OI = Operating Income
OE = Operating Expense

Points to be kept in mind:
- Provides absolute profitability
- Promotes goal congruence
- There is a greater probability that managers will be encouraged when acting in their own and company's best interest.

Limitation of RI
- Two companies different in size of sales/ assets should not be compared using RI because large division is more likely to earn a larger RI then a small division.

Economic Value added

Stakeholders can decide that the Company is generating or destroying their wealth

Measures based on residual wealth calculated by
Operating Profit- COC

The Formula for calculating EVA:

$$EVA = [\ NOPAT - (\text{Invested Captial} \times WACC)$$

Where : NOPAT = Net Operating Profit After Taxes
Invested Capital = ESH + R&S + PSH + Debt
WACC = Weighted Average Cost of Capital

Points to be kept in mind:
- If profit after interest and tax is given; add back interest (Net of Tax)
- Opening Capital employed should be taken
- Add back non cash item and accounting depreciation if included in operating profit.
- Deduct economic depreciation
- Accrued Tax: Add Back -Tax Paid: Deduct
- Deferred revenue expenditure (Advt., R&D) if debited in P/L should be added back.
- If previous year adjustment are not done for non cash item/ PFDD/ Deferred revenue exp. (R&D) then add back in opening capital employed also.

Limitation of EVA
- An absolute measure hence comparison of performance between enterprises of different size not possible.
- Largely based upon historical data.

How to increase EVA			
NOPAT	Increase	Increase more than Capital Employed	Keep as it is
Capital Employed:	Keep as it is	Increase	Remove unwanted capital employed

In EVA show

$$WACC = \left(\begin{matrix}\text{weight of}\\ \text{equity}\end{matrix} \times \begin{matrix}\text{cost of}\\ \text{equity}\end{matrix}\right) + \left(\begin{matrix}\text{weight}\\ \text{of debt}\end{matrix} \times \begin{matrix}\text{After tax}\\ \text{cost of}\\ \text{debt}\end{matrix}\right)$$

In EVA

EBIT
(-) Int.
EBT
(-) Tax
EAT
(+) Int.
(Net of Tax)
NOPAT

Chapter 9 Divisional Transfer Pricing

1 Three situations in case of TP

Situation 1. When there is Full idle capacity.

MINIMUM TRANSFER PRICE

Where Idle Capacity is more than Internal Demand

(Variable Cost
- Variable Selling Cost, if any)

MAXIMUM TRANSFER PRICE

(Market Price
- Variable Selling Cost, if any)
Or (Variable cost + Opportunity cost)
Compare With
(Substitute Value - Alteration Cost)
(Whichever is Lower)

Situation 2. When there is Zero Idle Capacity. ie. Fully occupied.

Where No Idle capacity exists

MINIMUM TRANSFER PRICE

(Market Price - Variable Selling Cost, if any) Or (Variable cost + Opportunity cost)

MAXIMUM TRANSFER PRICE

(Market Price - Variable Selling Cost, if any) Or (Variable cost + Opportunity cost)

Situation 3. When there is part idle capacity and part occupied capacity

Capacity is less than or equal to Internal Demand

MINIMUM TRANSFER PRICE

Average of the:

- Idle Capacity @ Variable Cost
- \+
- Capacity Sacrificed @ Variable Cost + Opportunity Cost

MAXIMUM TRANSFER PRICE

(Market Price - Variable Selling Cost, if any) Or (Variable cost + Opportunity cost)

Compare With

(Substitute Value - Alteration Cost)

Whichever is Lower

2 Proposals For Resolving Transfer Pricing Conflict

Conflict of interest between interests of individual divisions and the company can also be addressed by following the following systems for transfer pricing:

• Dual Rate Transfer Pricing System

The supplying division records transfer price by including a normal profit margin thereby showing reasonable revenue. The purchasing division records transfer price at marginal cost thereby recording purchases at minimum cost. This allows for better evaluation of each division's performance. It also improves co - operation between divisions, promoting goal congruence and reduction of sub -optimization of resources.

Drawbacks of Dual Pricing include:

- It can complicate the records, thereby may result in errors in the company's overall records.
- Profits shown by the divisions are artificial and need to be used only for internal evaluations.

• Two Part Transfer Pricing System

This pricing system is again aimed at resolving problems related to distortions caused by the full cost based transfer price.

Here, transfer price = marginal cost of production + a lump -sum charge (two part to pricing).

While marginal cost ensures recovery of additional cost of production related to the goods transferred, lump -sum charge enables the recovery of some portion of the fixed cost of the supplying division. Therefore, while the supplying division can show better profitability, the purchasing division can purchase the goods a lower rate compared to the market price.
Eg : Div A is training dept providing training to Div B & C. Div A incurrs Rs 10,00,000 fixed cost p.a. and Rs 750 variable cost for training.
TP = 750 + Some Portion of RS 10,00,000

Question — **GL Ltd** — **(4 divisions)**

Electric Div → Transfer 5000 units → Household Div

(Fully Utilised with external Demand)

@	Rs25
	(MP)
⊖	
VC	16
Contribution p.u.	9 p.u.

(70% Capacity)

TP @ 18/-

SP	: 130
⊖ Comp	: (50)
⊖ Switch	: (18)
⊖ OVC	: (40)
Contribution p.u.	: 22
Fixed Overhead p.u.	: 21

Impact of different TP on overall Profitability

If TP > 18 (or 25)... then it will not be affordable to the Household division.

If TP > 18

- Electric Div → Electric Div → 5000 Switches × (25 - 16 MP VC)
- But HH division will be closed down

Contribution = 5000 x 9
= 45000/-

3 TP to be charged by Elec div. if it is Fully Occupied.

Since Elec. division is fully occupied, so it will transfer its goods and services @ MP.
According, any price bellow Rs 25/- (i.e its MP) will not be acceptable.
Also, since Elec Div. is a profit centric division (based on ROI) so it will not transfer below its MP as it is fully occupied.

If TP < 18

Electric Div	HH Div
TP = 18	will continue
⊖ VC = 16	↓
C = 2	Couth 22 p.u.
x 5000 u	@ 5000 u
10,000	Rs 110,000

→ Overall co. benefit = 120,000 Rs
⊖ Already Earned = 45,000 Rs
Rs. 75000
÷ 5000 u
= Rs 15 p.u.

Additional Benefit to the Company will be

Range of T.P.

Min TP (Transferor) (Isko kitna minimum chahiye hi chahiye)

Electrical
Full Occupied @
min TP = MP
= 25/-

Max TP (Transferee) (Yeh max kitna de sakta hai)

Household
SP : 130
⊖ Comp : (50)
⊖ OVC : (40)
Max (40)

Range = 25 - 40

What will be the Maximum transfer price if in above scenario substitute is available to Transferee at

Maximum : TP	@ 35/-	@ 45/-
MP	25	25
(or) Substitule Value	35	45
(or) (SP - All variable cost) (excluding T.P)	40	40

MAX TP : Substitute value = Rs 35/-
(SP - All variable cost) = Rs 40/-

Hint : Centre Value

Chapter 12 Standard Costing

1 ANALYSIS OF MATERIAL VARIANCE

Material Cost Variance
(Standard Material Cost - Actual Material Cost)
(SQ x SR - AQ x AR)
{[Standard Quantity of input for Actual output(SQ) x Standard rate of input (SR)]
-[Actual Quantity consumed (AQ) x Actual rate of input (AR)]}

Material Price Variance
(Standard rate -Actual rate) x Actual quantity of input
(SR - AR) x AQ

Material Usage Variance
(Standard quantity of input for actual output - Actual input) x Standard rate
(SQ - AQ) x SR

Derivation of Sub Usage Variance
Adding & Subtracting SMQ from both sides
(SQ - SMQ + SMQ - AQ) x SR
(SQ - SMQ) x SR + (SMQ - AQ) x SR

Material Sub Usage Variance
(When the total quantity of standard and actual is different)
(SQ - SMQ) x SR

Material Mix Variance
(When the ratio of standard and actual quantity is different)
(SMQ - AQ) x SR

Where, SMQ = Standard Mix Quantity
Calculated as Total of Actual Quantity in Standard Ratio

Variance Table = If, SM > AM = (F)
SR > AR = (F)
SM > SMM = (F)
SMM > AM = (F)

2 ANALYSIS OF LABOUR VARIANCE

Labour Cost Variance
(Standard Labour Cost - Actual Labour cost)
(Standard hours for Actual output x Standard Rate per labour hour - Actual Hours paid x Actual Rate)

Labour Efficiency Variance
[(Std hours for Actual Output-Actual hrs) x Std Rate per labour hr]

Derivation of Sub Efficiency Variance
(SH - AH) x SR
(SH - SMH + SMH - AH) x SR
(SH - SMH) x SR + (SMH - AH) x SR

Idle Time Variance
(Actual Hrs Paid - Actual Hrs Worked) x Standard Rate per labour hr

(Always Adverse)

Labour Rate Variance
[(Std Rate - Actual Rate) x Actual Labour hrs PAID]
(SR - AR) x AH

Labour Sub Efficiency Variance
(When **TOTAL** of Standard hours and Actual hours is different)
(SH - SMH) x SR
Adding & Subtracting SMH form both sides
(SH - SMH + SMH - AH) x SR

Labour Mix Variance
(When **RATIO** of Standard and Actual hours is different)
(SMH - AH) x SR

Where, SMH = Standard Mix Hours
SMH = Total of Actual Hours in Standard Ratio

Total Hour is same for Labour, Variable and Fixed Overheads

Idle Time
Actual hours paid (gross hours)
- Actual hours worked (net hours)

Variance Table = If, SH > AH = (F)
SR > AR = (F)
SH > SMH = (F)
SMH > AH = (F)

3 ANALYSIS OF VARIABLE OVERHEAD VARIANCES

Variable Cost Variance

(Std Budgeted Variable OH - Actual Variable OH)

(Std Variable OH Rate p.u x Actual output - Actual Variable Rate p.u x Actual Output)

(Std Hours for Actual x Std VOH Rate per hr - Actual Hrs x Actual Variable Rate / Hrs)

Variable OH Expenditure Variance

(Std variable OH Rate per hour - Actual Variable OH Rate per hr) x Actual hrs worked

Variable OH Efficiency Variance

(Std Hours for Actual Production - Actual hrs worked x Std Variable OH Rate per hour

Total Hour is same for Labour, Variable and Fixed Overheads

4 ANALYSIS OF FIXED OVERHEAD VARIANCES

Fixed Overhead Cost Variance

(Absorbed Fixed Overheads - Actual fixed overhead)

(Actual production x RECOVERY RATE P.U.- Actual Fixed Overheads)

(Actual Production x Standard Hours per unit x Standard Rate Per Hour - Actual Fixed Overhead)

Fixed OH Expenditure Variance

Budgeted Fixed OH - Actual Fixed OH

Fixed OH Volume Variance

(Budgeted Production - Actual Production) x RR p.u.

(Budgeted Prodn - Actual Prodn) x Std hours x RR per hr.

(Budgeted Prodn x Std Hours) - (Actual Prdn x (Std hr)] x RR/hr.

(Budgeted hrs for Budget Prdn - Std Hrs. for Actual Prdn) x RR per hour

(ALWAYS CALCULATED IN TERM OF HOURS)

Fixed OH Capacity Variance

(Budgeted Hrs - Actual hours) x RR per hour

Fixed OH Efficiency Variance

(Std. hours for Actual production - Actual hours) x RR per hour

Fixed OH Calender Variance

(Budgeted days - Actual days) x RR per days

Fixed OH Revised

Capacity Variance (Budgeted hours for Actual days - Actual Hours) x Recovery Rate per hour

5 Overhead variances arise due to difference between Actual overhead and Absorbed overheads.

Variable Overhead	Fixed Overhead	
Std. Variable Ohd rate = $\frac{\text{Std Voh}}{\text{Std Hours}}$	RR/u : $\frac{\text{Budgeted Fixed Overheads}}{\text{Budgeted Production}}$	RR/u : Std hours p.u. x RR/ hour
Actual Variable ohd rate = $\frac{\text{Actual Voh}}{\text{Act hours}}$	RR/hr : $\frac{\text{Budgeted Fixed Overheads}}{\text{Budgeted Hours}}$	Std hours p.u. : $\frac{\text{Budgeted Hours}}{\text{Budgeted Production}}$

6 Variance Table

Variances Table= Budget Fixed Overhead > Actual Fixed Overheads = (F)
Budget Production > Actual Production = (A)
Budget Hours (capacity) > Actual Hours = (A)
Standard Hours for Actual Production > Actual Production = (F)
Standard Days > Actual Days = (A)

Following ratios are derived from the formulas of Fixed OH Variance:

1. Efficiency Ratio = $\frac{\text{Standard hours for Actual Production}}{\text{Actual hours for Actual Production}} \times 100$ [Activity Ratio X Capacity Ratio]

2. Capacity Ratio = $\frac{\text{Actual hours}}{\text{Budgeted hours}} \times 100$ [Efficiency Ratio X Activity Ratio]

3. Activity Ratio = $\frac{\text{Standard hours for Actual Production}}{\text{Budgeted hours for Budgeted Production}} \times 100$ [Efficiency Ratio X Capacity Ratio]

7 SALES VARIANCE

a) Sales Variances by Turnover Method

Sales Value Variance:
(Budget Sales Value - Actual Sales Value)

(Budgeted Sales Value - Actual Sales)
(Budgeted Sales Qty x Budgeted Selling Price)
- (Actual Sales Qty x Actual Selling Price)

Sales Price Variance
[(Budgeted SP - Actual SP) x Actual sales qty]

Sales Volume Variance
[(Budgeted Sales Qty - Actual sales qty) x Budgeted SP]

Sales Quantity Variance
(Budgeted sales qty - Std Sales mix qty)
x Budgeted SP
(BSQ - SSMQ) x BSP

Sales Mix Variance
(Std Sales mix qty - Actual sales qty)
x Budgeted SP
(SSMQ - ASQ) x BSP

b) Sales Variances by Margin Method

Sales Margin Value Variance:
(Budgeted Margin - Actual Margin)
[(Budgeted Sales Qty x Budgeted Margin)
- (Actual Sales Qty x Actual Margin)]

Sales Margin Price Variance
[(Budgeted margin - Actual margin)
x Actual sales qty]

Sales Margin Volume Variance
[(Budgeted sales qty - Actual sales qty)
x Budgeted margin]

Sales Margin Quantity Variance
[(Budgeted sales qty - Std Sales mix qty)
x Budgeted margin]
(BSQ - SSMQ) x BM

Sales Margin Mix Variance
[(Std Sales mix qty - Actual sales qty)
x Budgeted margin]
(SSMQ - ASQ) x BM

*BM = BSP - BC
AM = ASP - BC

8 SALES MARGIN QUANTITY VARIANCE

Sales Margin Market Size Variance	
Budgeted Market Size	xx
(-) Actual Market Size	xx
	xx
X Our share in Market	
X Budgeted Margin / Budgeted selling price	

Sales Margin Market Share Variance
(ASQ - Industry Sales as per our share) x Budgeted Margin / Budgeted selling price

9 Planning & Operation When there is only 1 input

Price Variance

Planning
=(Original Standard Rate - Revised Standard Rate) x Revised Standard Quantity

Operational
= (Revised Standard Rate - Actual Standard Rate) x Actual Quantity

Usage Variance

Planning
= (Original Standard Quantity - Revised Standard Quantity) x Original Standard Rate

Operational
= (Revised Standard Quantity - Actual Quantity) x Revised Standard Rate

10 Planning & Operations When there is more than 1 input

(Planning) Price Variance

Uncontrollable
= (Original Standard Rate - Actual Input Rate) x Revised Standard Quantity

Controllable
= (Actual Input Rate - Revised Input rate) x Revised Standard Quantity

Operational

Material Price Variance
= (Revised Standard Rate - Actual Rate) x Actual Quantity

Material Usage Variance
= (Revised Standard Quantity - Actual Quantity) x Revised Standard Rate